Cardiovascular Disease Prediction: Cost-Sensitive Screening with Threshold Optimization

Author: Md. Anwarul Karim, MS in Data Science and Analytics, East West University

Code: `cvd_pipeline.py` (Python, scikit-learn) | **Data:** `CVD_cleaned.csv`

Abstract

This project builds a screening model for self-reported cardiovascular disease (CVD) using a large public health survey (N = 308,754 after cleaning, 8.09% positive). Because the classes are heavily imbalanced, accuracy is a misleading metric: a model that always predicts "no disease" would be 91.9% accurate and detect no one. The pipeline therefore optimizes recall, PR-AUC and probability calibration. Four models were compared under identical leak-free preprocessing, and the best-performing one (gradient boosting) was calibrated with isotonic regression. Its decision threshold was chosen from out-of-fold training predictions to target 80% recall, without using the test set. On a held-out test set of 61,751 patients (4,993 with CVD), the final model reached **ROC-AUC 0.836, PR-AUC 0.301 (baseline 0.081), recall 80.3%, specificity 72.5%, precision 20.4%** and a Brier score of 0.064. Bootstrap 95% confidence intervals and an ablation study are reported. Age and self-rated general health are the dominant predictors.

1. Problem and Motivation

Screening for cardiovascular disease has asymmetric costs: missing a true case is usually worse than sending a healthy person for a follow-up test. Standard classifiers optimized for accuracy ignore this and default to the majority class. This project asks:

1. How well can a handful of survey variables (demographics, comorbidities, lifestyle) identify people with CVD?
2. Can a decision threshold be chosen to reach a target sensitivity honestly, without tuning on test data?

Are the predicted probabilities reliable enough to interpret as risk?

What the data can and cannot answer. The data come from the CDC Behavioral Risk Factor Surveillance System (BRFSS), a self-reported telephone survey, not clinical records. The target is whether a respondent was ever told by a professional that they had coronary heart disease or a heart attack. The model therefore **detects existing, self-reported disease**. It does not predict future cardiovascular events.

2. Data

Item	Value
File	`CVD_cleaned.csv`
Raw shape	308,854 rows × 19 columns
Missing values	0
Target	`Heart_Disease` (Yes/No)
Class balance after cleaning	8.09% positive (24,965 of 308,754)

Variable groups

- **Ordinal:** `General_Health` (5 levels), `Age_Category` (13 bands), `Checkup` (5 levels)
- **Nominal:** `Diabetes` (4 levels)
- **Binary:** `Exercise`, `Skin_Cancer`, `Other_Cancer`, `Depression`, `Arthritis`, `Sex`, `Smoking_History`
- **Numeric:** `BMI`, `Alcohol_Consumption`, `Fruit_Consumption`, `Green_Vegetables_Consumption`, `FriedPotato_Consumption`
- **Dropped:** `Height_(cm)` and `Weight_(kg)`, because BMI already captures them

3. Data Cleaning Decisions

3.1 Biologically grounded BMI filter. A conventional 1.5×IQR outlier filter across BMI and the four dietary variables would remove 122,077 patients (39.5% of the data) and cap BMI at about 43.3, discarding the severely obese patients who are an important risk group. Instead I removed only biologically implausible BMI values (below 12 or above 75). Only **100 rows** were removed (94 non-CVD, 6 CVD).

Filter State	Rows
Before filtering	308,854
After filtering	**308,754**

3.2 Diabetes categories. The four survey answers were renamed into clean clinical states: No, Pre-Diabetes, Gestational Diabetes, Yes.

3.3 Duplicates. 100 exact duplicate rows remain (after height and weight were dropped). They were kept because in survey data identical answers can come from different people. They could appear in both the train and test sets, which is noted as a limitation.

4. Exploratory Data Analysis

4.1 Why prevalence rates and not counts

Because 91.9% of respondents are disease-free, raw count plots always show most cases among healthy-looking groups (for example, people who exercise). This is the base-rate fallacy. All plots in this project show **within-group prevalence**, P(CVD | feature), the percentage of each group with the disease.

4.2 Prevalence by group (full tables, N = 308,754)

Self-reported general health

Group	n	CVD prevalence
Excellent	55,943	2.0%
Very Good	110,380	4.4%
Good	95,343	9.1%
Fair	35,776	19.0%
Poor	11,312	31.8%

Age category

Group	n	CVD prevalence
18-24	18,676	0.5%
25-29	15,489	0.7%
30-34	18,423	1.1%
35-39	20,598	1.3%
40-44	21,579	2.0%

Group	n	CVD prevalence
45-49	20,958	3.2%
50-54	25,080	4.7%
55-59	28,042	7.1%
60-64	32,408	9.3%
65-69	33,430	11.4%
70-74	31,100	14.7%
75-79	20,703	18.1%
80+	22,268	21.8%

Diabetes

Group	n	CVD prevalence
No	259,074	6.1%
Gestational diabetes	2,645	3.6%
Pre-diabetes / borderline	6,891	11.5%
Yes	40,144	20.9%

Other factors

Factor	Group	n	CVD prevalence
Arthritis	No	207,727	5.2%
Arthritis	Yes	101,027	14.1%
Smoking history	No	183,537	5.7%
Smoking history	Yes	125,217	11.6%

Factor	Group	n	CVD prevalence
Exercise	No	69,415	13.0%
Exercise	Yes	239,339	6.7%
Depression	No	246,892	7.6%
Depression	Yes	61,862	9.9%
Sex	Female	160,140	6.2%
Sex	Male	148,614	10.1%
Other cancer	No	278,884	7.3%
Other cancer	Yes	29,870	15.8%
Skin cancer	No	278,762	7.3%
Skin cancer	Yes	29,992	15.6%

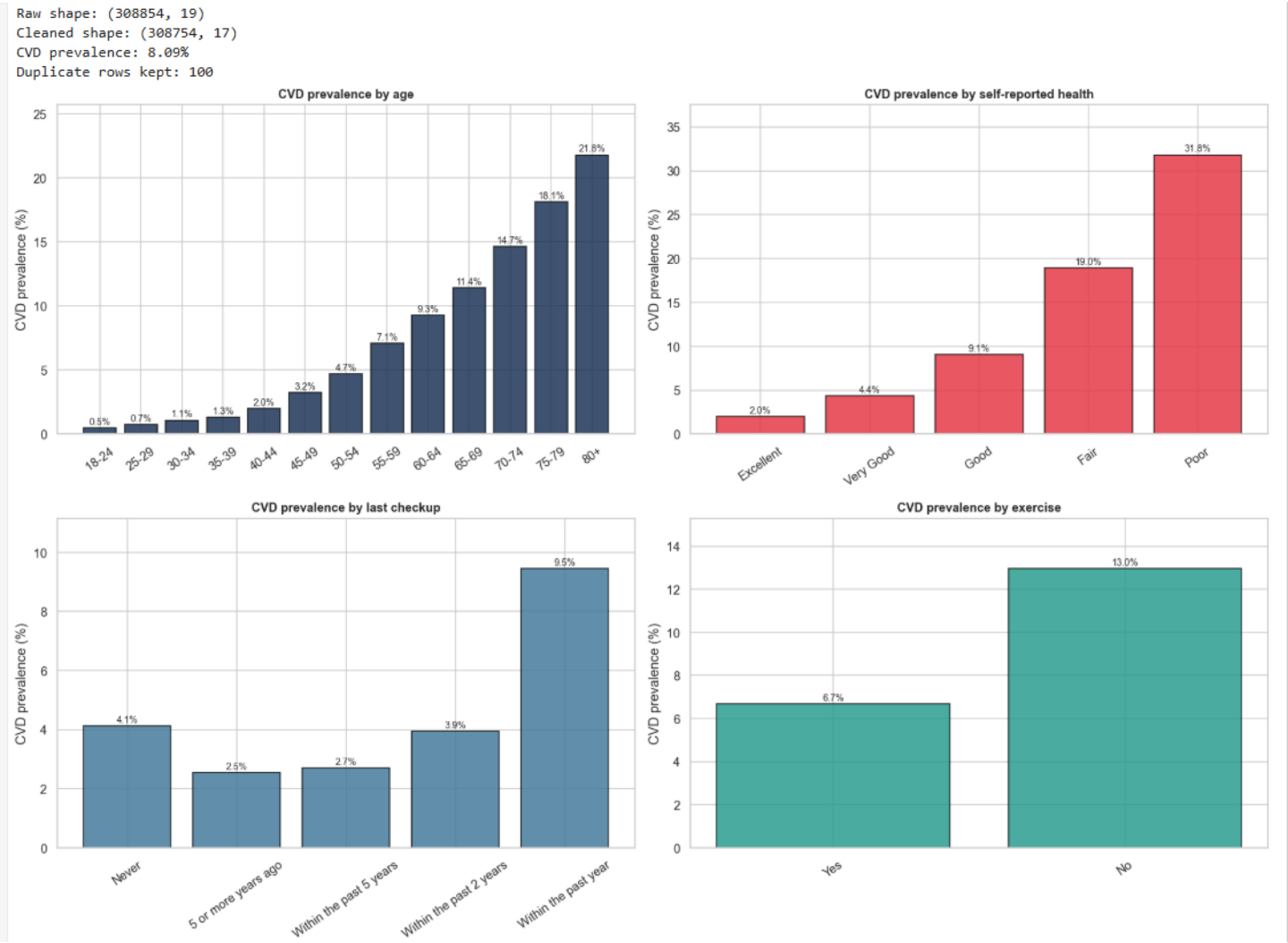
Last checkup

Group	n	CVD prevalence
Within the past year	239,295	9.5%
Within the past 2 years	37,202	3.9%
Within the past 5 years	17,436	2.7%
5 or more years ago	13,415	2.5%
Never	1,406	4.1%

BMI

Statistic	Value
Mean / SD	28.61 / 6.45
Min / median / max	12.02 / 27.44 / 74.68

BMI quartile	CVD prevalence
Q1 (12.0–24.2)	6.1%
Q2 (24.2–27.4)	7.7%
Q3 (27.4–31.8)	8.9%
Q4 (31.8–74.7)	9.7%



[figures: eda_prevalence_1.png]

4.3 Observations

- **Risk climbs steadily with poorer self-rated health and with age.** Prevalence rises about 16-fold from "Excellent" to "Poor" health, and from 0.5% at ages 18-24 to 21.8% at 80+. Risk passes 7% at ages 55-59 and keeps rising after that.
- **Diabetes, arthritis, smoking history and other cancers** each roughly double risk or more (diabetes is about 3.4× the non-diabetic rate).
- **Exercise is associated with lower prevalence** (6.7% vs 13.0%), but this is confounded by age and health. The model's later importance scores support that (see Section 8).
- **BMI shows only a modest gradient** (6.1% to 9.7% from the lowest to the highest quartile). That is weaker than the other factors.
- **The checkup pattern is counterintuitive.** People who saw a doctor in the past year have the highest prevalence (9.5%). This is reverse causation: people with heart disease visit doctors more often, so checkup frequency is a consequence of disease and not a risk factor.

5. Methods

5.1 Encoding

- `General_Health`, `Checkup` and `Age_Category` use explicit ordinal maps in true clinical order. Alphabetical label encoding would have scrambled them.
- Binary variables are coded 0/1. `Diabetes` is one-hot encoded.
- An assertion confirms that no category went unmapped.

5.2 Train/test split

A stratified 80/20 split with `random_state = 42`.

Set	Patients	CVD prevalence
Train	247,003	8.086%
Test	61,751 (4,993 positives)	8.086%

5.3 Leak-free preprocessing

A `ColumnTransformer` inside a scikit-learn `Pipeline` fits the scaler and encoder on the training data only:

- `StandardScaler` on BMI and the four dietary variables
- One-hot encoding of `Diabetes` (first level dropped)
- All ordinal and binary columns passed through unchanged

5.4 Models

Model	Key settings
Logistic Regression	`class_weight="balanced"`, C = 1.0
Decision Tree	`max_depth=8`, `min_samples_leaf=50`, balanced
Random Forest	200 trees, `max_depth=10`, `min_samples_leaf=25`, balanced
HistGradientBoosting	250 iterations, `learning_rate=0.05`, `max_depth=6`, `min_samples_leaf=30`, balanced

Balanced class weights make a missed positive case cost more than a false alarm during training.

5.5 Calibration and threshold selection

1. The gradient-boosting model was wrapped in `CalibratedClassifierCV` (isotonic regression, 3 internal folds). Isotonic calibration turns scores into probabilities that match observed risk.
2. Calibrated predictions were generated for the **training data** using 5-fold out-of-fold prediction.
3. Using the precision-recall curve on those training predictions, I chose the highest threshold that still gives at least 80% recall. The result was **τ* = 0.0872**.

The calibrated model was refit on the full training set. The **test set was used only once**, to report results.

This avoids the common mistake of choosing the threshold on the test data, which makes the reported recall optimistic.

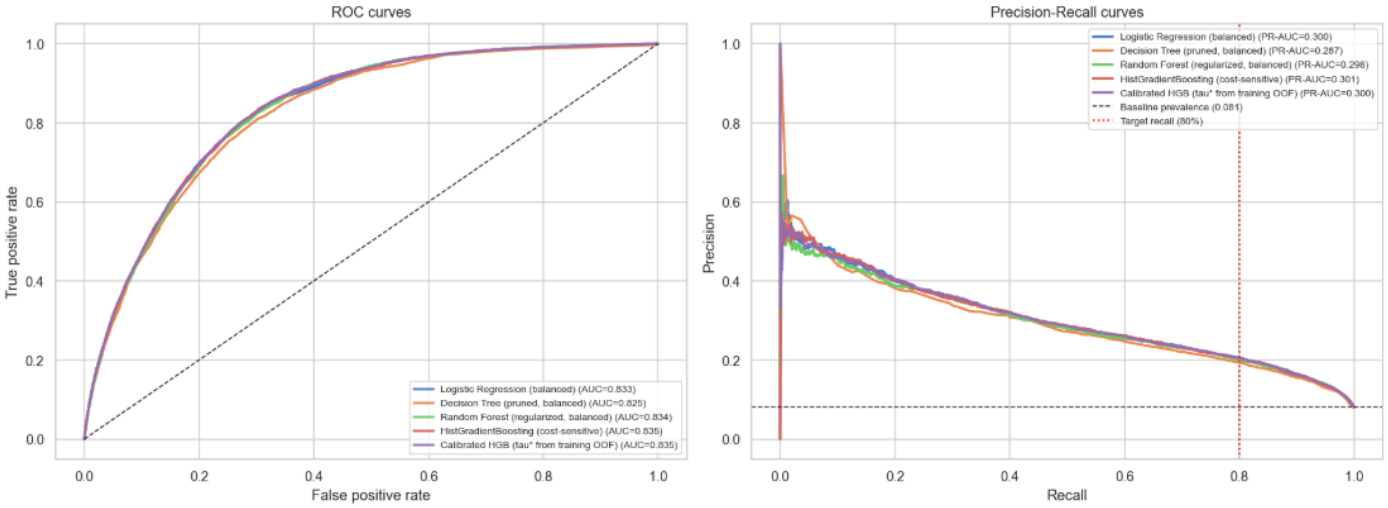
6. Results

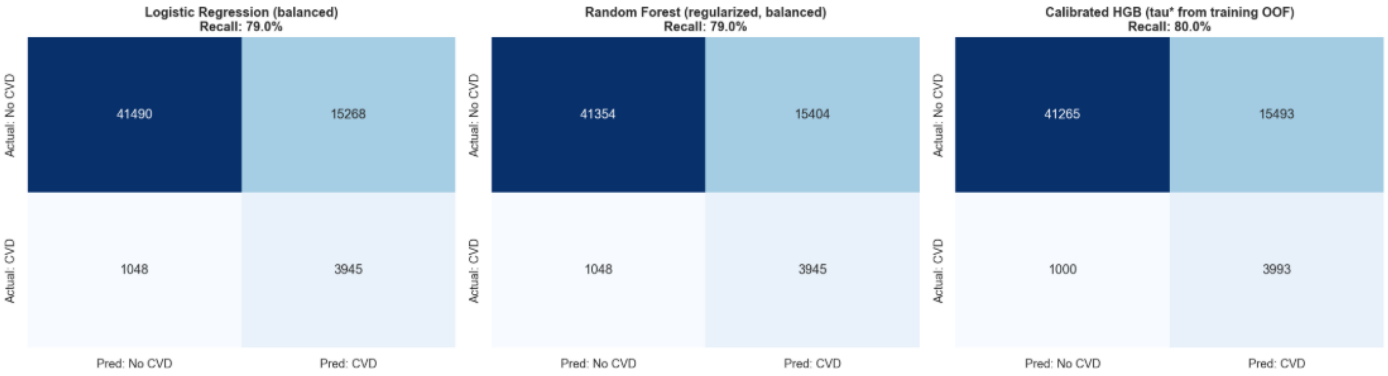
6.1 Test-set performance (N = 61,751; 4,993 positives)

Model	Thresh	ROC-AUC	PR-AUC	Recall	Spec	Prec	F1	F2	Acc	Brier
Logistic Regression (balanced)	0.50	0.8331	0.2995	0.7901	0.7310	0.2053	0.3260	0.5034	0.7358	0.1726
Decision Tree (pruned, balanced)	0.50	0.8247	0.2871	0.7951	0.7108	0.1947	0.3129	0.4918	0.7176	0.1763

Model	Thresh	ROC-AUC	PR-AUC	Recall	Spec	Prec	F1	F2	Acc	Brier
Random Forest (regularized, balanced)	0.50	0.8338	0.2955	0.8057	0.7147	0.1990	0.3192	0.5005	0.7221	0.1700
HistGradientBoosting (cost-sensitive)	0.50	0.8352	0.3006	0.8065	0.7209	0.2027	0.3239	0.5054	0.7278	0.1708
Calibrated HGB, τ* from training OOF	**0.0872**	**0.8356**	**0.3008**	**0.8033**	**0.7246**	**0.2042**	**0.3256**	**0.5063**	**0.7310**	**0.0640**

```
=== Bootstrap 95% CI (final model) ===
ROC-AUC      0.8356  [0.8309, 0.8402]
PR-AUC       0.3006  [0.2892, 0.3130]
Recall       0.7999  [0.7896, 0.8102]
Specificity  0.7271  [0.7239, 0.7308]
Precision    0.2048  [0.1991, 0.2106]
```





[figure: roc_curves.png] [confusion_matrices.png]

6.2 Confusion matrices

Model	True neg	False pos	False neg	True pos
Logistic Regression	41,490	15,268	1,048	3,945
Decision Tree	40,342	16,416	1,023	3,970
Random Forest	40,567	16,191	970	4,023
HistGradientBoosting	40,916	15,842	966	4,027
Calibrated HGB (τ* = 0.0872)	**41,127**	**15,631**	**982**	**4,011**

6.3 What this means in practice

For the final model, out of 61,751 patients:

- **19,642 (31.8%)** are flagged as high risk. Of these, 4,011 truly have CVD and 15,631 do not.
- **982 cases (19.7%) are missed.**
- About **one in five flagged patients has CVD** (precision 20.4%), so roughly five people must be followed up to find one case.
- Overall accuracy is 73.1%, compared with 91.9% for always predicting "no". The drop is intentional: it buys the extra 80 percentage points of recall.

6.4 Interpreting the model comparison

7. Uncertainty: Bootstrap 95% Confidence Intervals

The test set was resampled 500 times with replacement, using the final calibrated model at τ* = 0.0872.

Metric	Mean	95% CI
ROC-AUC	0.8357	0.8310 – 0.8403
PR-AUC	0.3011	0.2896 – 0.3140
Recall	0.8035	0.7927 – 0.8139
Specificity	0.7247	0.7213 – 0.7284

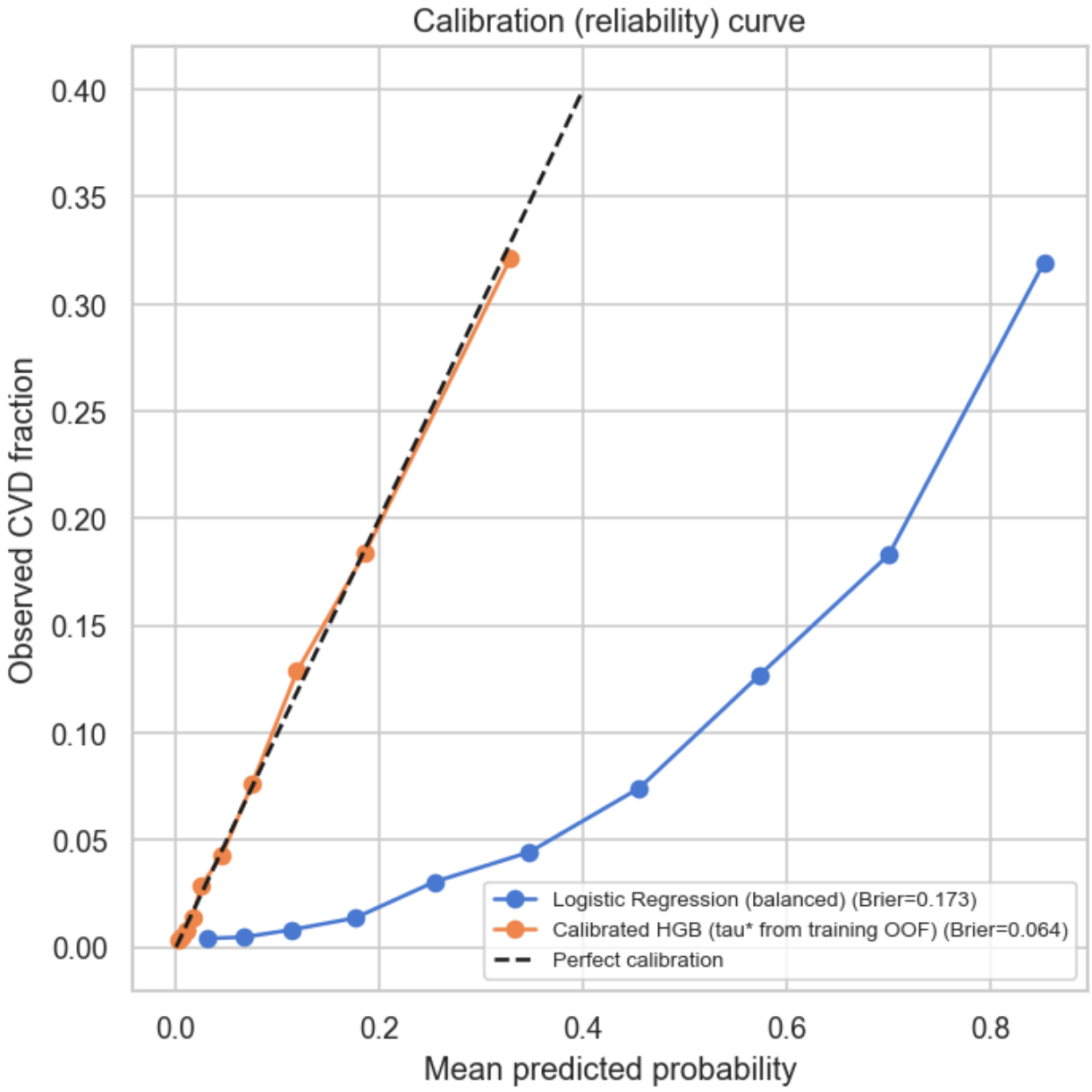
Metric	Mean	95% CI
Precision	0.2041	0.1985 – 0.2095

The 80% recall target is therefore an estimate of about **79–81%**, not a guarantee. It may differ on new populations.

8. Calibration

Isotonic calibration lowered the Brier score from 0.1708 (uncalibrated cost-sensitive HGB) to **0.0640** for the final model. Balanced class weights distort probabilities by design, so the uncalibrated models' scores cannot be read as risks. The calibrated model's predicted and observed risks agree closely in every decile:

Decile	Mean predicted risk	Observed CVD fraction
1	0.003	0.003
2	0.005	0.005
3	0.009	0.008
4	0.016	0.013
5	0.025	0.028
6	0.045	0.042
7	0.075	0.076
8	0.117	0.128
9	0.185	0.184
10	0.328	0.320



[figure: calibration_curve.png]

9. Feature Importance (Permutation, Test ROC-AUC)

Rank	Feature	Mean drop in ROC-AUC	Std
1	Age_Category	0.1233	0.0010
2	General_Health	0.0532	0.0010
3	Sex	0.0221	0.0012

Rank	Feature	Mean drop in ROC-AUC	Std
4	Diabetes	0.0077	0.0002
5	Smoking_History	0.0054	0.0004
6	Arthritis	0.0035	0.0002
7	Checkup	0.0032	0.0006
8	Depression	0.0017	0.0002
9	Alcohol_Consumption	0.0014	0.0003
10	Other_Cancer	0.0007	0.0001
11	BMI	0.0005	0.0003
12	Skin_Cancer	0.0005	0.0001
13	FriedPotato_Consumption	0.0004	0.0001
14	Green_Vegetables_Consumption	0.0002	0.0002
15	Fruit_Consumption	0.0002	0.0001
16	Exercise	0.0001	0.0000

Reading this table:

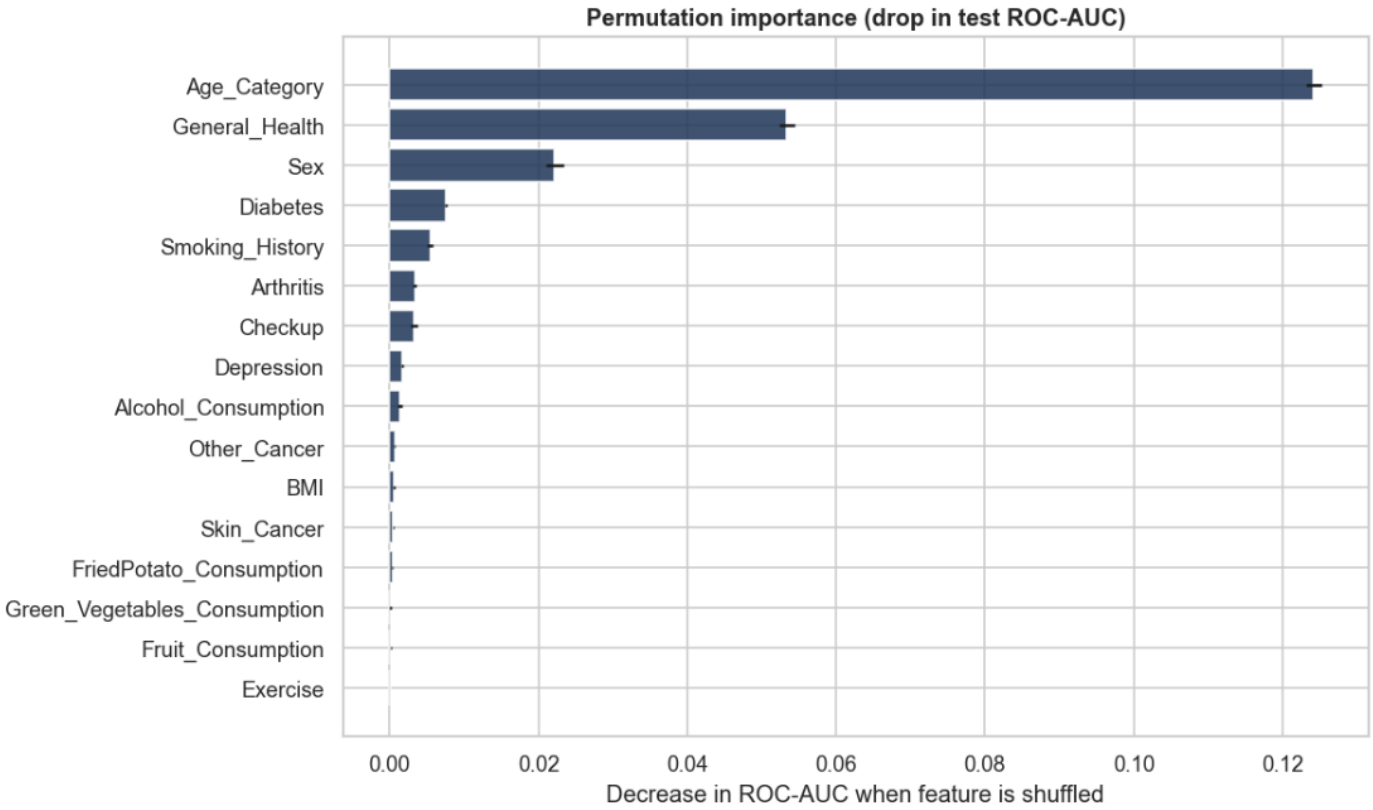
- **Age is by far the dominant predictor**, more than twice as important as the next feature.
- **BMI, exercise and the diet variables contribute almost nothing** once age and health are known. Their raw correlations with CVD (Section 4) are largely explained by the stronger variables.
- Keeping severely obese patients in the data was still the right data-integrity choice, but it did not change predictive performance much.
- Importance reflects predictive contribution, not causation.

10. Ablation: Removing Self-Rated Health

Self-rated general health could be a proxy for illness that the person already knows about. To check how much the model relies on it, I removed `General_Health` and retrained the gradient-boosting model.

Model	ROC-AUC	PR-AUC
With `General_Health`	0.8352	0.3006
Without `General_Health`	0.8174	0.2643

ROC-AUC drops by about 0.018 and PR-AUC by about 0.036. Self-rated health helps, but the model still performs well without it, so it is not carrying the result on its own.



Feature	Mean	Std
Age_Category	0.1242	0.0011
General_Health	0.0534	0.0010
Sex	0.0222	0.0012
Diabetes	0.0076	0.0002
Smoking_History	0.0054	0.0004
Arthritis	0.0034	0.0002
Checkup	0.0033	0.0005

[figure: permutation_importance.png]

11. Discussion

- **The framing worked.** Optimizing for recall and choosing the threshold deliberately turned a model that would find nobody into one that finds about 80% of cases.
- **The ceiling is the data, not the algorithm.** Four very different models land within about 0.01 AUC of each other. With only 16 coarse self-reported variables, more complex models are unlikely to help much.

- **The threshold is a policy choice.** An 80% recall target is a design choice. Whether the false-positive cost (about 15,600 follow-ups here) is acceptable depends on how cheap the next step is. A lower or higher recall target would trade precision accordingly.
- **Calibration makes the model more usable.** Clinicians can read the output as a probability and reason about a risk cutoff, not an arbitrary score.

12. Limitations

1. **Self-reported data.** Neither the inputs nor the label are clinically verified, and recall bias is possible.
2. **Detection, not prediction.** The label is "ever diagnosed", so the model identifies existing disease and does not forecast future events.
3. **No survey weights.** BRFSS sampling weights were not applied, so prevalence figures describe this sample, not the national population.
4. **No external validation.** All results come from one random split of a single dataset. Performance on another population or year is unknown.
5. **Reverse causation.** Variables such as recent checkups and exercise levels may be consequences of disease, not causes of it.
6. **Modest precision.** About 80% of flagged patients do not have CVD, so this only makes sense as a low-cost first-stage screen.
7. **Duplicates.** 100 duplicate rows were left in and may appear in both the training and test sets.
8. **Fairness not evaluated.** Performance by sex, age group and other subgroups was not analyzed.

13. Conclusion

A leak-free, cost-sensitive, calibrated gradient-boosting model reaches ROC-AUC 0.836 and PR-AUC 0.301 (3.7× the 8.1% baseline). At a threshold chosen without touching the test set, it detects about 80% of CVD cases at about 72.5% specificity. Probabilities are well calibrated (Brier 0.064), results are accompanied by confidence intervals, and age and self-rated health drive most of the signal. The main weaknesses are the self-reported data and the lack of external validation. These are properties of the data source, not of the pipeline.

Possible next steps: apply BRFSS survey weights; validate on a different BRFSS year; evaluate subgroup fairness; compare against a logistic regression with interaction terms; and add cost-based threshold analysis using realistic follow-up costs.

Appendix A: Reproducibility

- `Python` with `pandas`, `numpy`, `scikit-learn (≥ 1.2)`, `matplotlib`, `seaborn`
- Random seed 42 throughout
- Run `cvd_pipeline.py` (or the notebook version) with `CVD_cleaned.csv` in the same folder
- Figures are saved to `figures/` and the results table to `figures/results_table.csv`